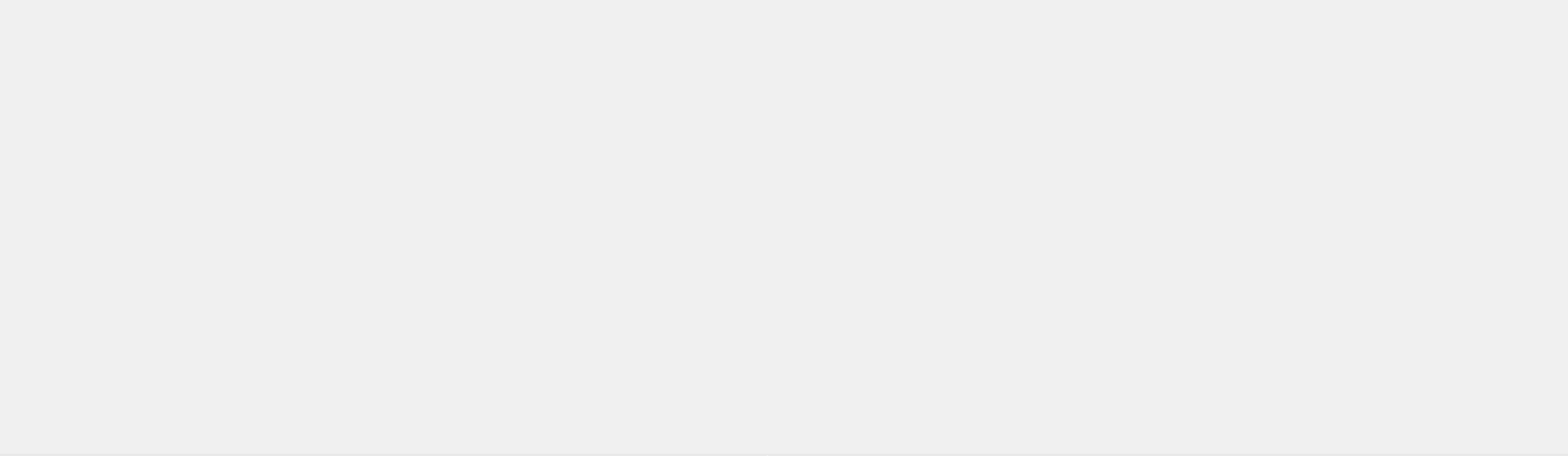




Democracy Dies in Darkness

Moon helium deal is biggest purchase of natural resources from space

The feasibility of moon mining is not yet proven, but the future of supercomputing may depend on the ability to extract Helium-3 from the lunar surface.

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By Kasha Patel

We haven't mined on the moon yet, but companies are already buying its resources.

Finnish tech firm Bluefors, a maker of ultracold refrigerator systems critical for quantum computing, has purchased tens of thousands of liters of Helium-3 from the moon — spending “above \$300 million” — through a commercial space company called Interlune. The agreement, which has not been previously reported, marks the largest purchase of a natural resource from space.

Interlune, a company founded by former executives from Blue Origin and an Apollo astronaut, has faced skepticism about its mission to become the first entity to mine the moon (which is legal thanks to a 2015 law that grants U.S. space companies the rights to mine on celestial bodies). But advances in its harvesting technology and the materialization of commercial agreements are gradually making this undertaking sound less like science fiction.

Bluefors is the third customer to sign up, with an order of up to 10,000 liters of Helium-3 annually for delivery between 2028 and 2037.

The deal is “the strongest signal we can figure out that says there is a demand for Helium-3,” said Rob Meyerson, CEO of Interlune and former president of Blue Origin. (Blue Origin founder Jeff Bezos owns The Washington Post.)

This isn't balloon helium. Helium-3 is lighter than the Helium-4 gas featured at birthday parties. It's also much rarer on Earth. But moon rock samples from the Apollo days hint at its abundance there.

Interlune has placed the market value at \$20 million per kilogram (about 7,500 liters).

“It's the only resource in the universe that's priced high enough to warrant going out to space today and bringing it back to Earth,” Meyerson said.

Why supercomputing may require moon mining

The Bluefors deal comes ahead of a highly anticipated explosion of quantum computing, which could drastically improve the ability to solve some of the largest problems in the universe. The advanced computers can help identify novel approaches to treat diseases, discover habitable planets, produce better financial analysis or, more dangerously, crack access to sensitive information.

To understand the computer's potency, take this common thought experiment: A salesperson must travel to 22 cities, visiting each city only once and returning to the first city in the most efficient route. It sounds simple, but a laptop computer would take about 1,000 years to figure out the best course. A quantum computer could one day compute the solution in minutes, or maybe even seconds.

These next-level machines rely on particles called qubits to process information in new ways. But heat, even in small doses, can cause qubits to produce errors. That's where Helium-3 comes in.

Bluefors makes the cooling technology that allows the computer to operate — producing chandelier-type structures known as dilution refrigerators. Their fridges, used by quantum computer leader IBM, contain a mixture of Helium-3 and Helium-4 that pushes temperatures below 10 millikelvins (or minus-460 degrees Fahrenheit).

“It's like 200 times colder inside a Bluefors fridge than outer space, which is pretty amazing,” said Ingela Waismaa, director of marketing and communications at Bluefors. “You need to go down to 7 millikelvin so that the qubit can live in that environment.”

The quantum computing scene is in its early stages, said Gary Lai, a co-founder and chief technology officer of Interlune, who was previously the chief architect at Blue Origin.

Existing quantum computers have been built with more than a thousand qubits, he said, but a commercial system or data center would need a million or more. That could require perhaps thousands of liters of Helium-3 per quantum computer.

“They will need more Helium-3 than is available on planet Earth,” Lai said. Most Helium-3 on Earth, he said, comes from the decay of tritium (an isotope of hydrogen) in nuclear weapons stockpiles, but between 22,000 and 30,000 liters are made each year.

The moon could bridge the gap. As the sun sends out a stream of charged particles in space, it has showered the moon with Helium-3 since its birth. (Whereas Earth’s magnetic field deflects much of the Helium-3 from the sun.)

“We estimate there’s more than a million metric tons of Helium-3 on the moon,” Meyerson said. “And it’s been accumulating there for 4 billion years.”

Now, they just need to get it.

How to mine Helium-3 on the moon

The planned mining, or harvesting, of Helium-3 on the moon would involve four main steps.



First, Interlune would dig up some of the lunar surface and sort smaller particles from larger ones using centrifugal force. Working with the fine particles, the company would then release a mix of gases trapped in bubbles on the outer surface — similar to popping bubble wrap.

The next step would involve separating the Helium-3 from Helium-4 and other gases for shipment back to Earth. Once the process is finished, Interlune would return the excavated dirt back to the lunar surface — leaving an impression more like tilled farmland than a mining site.

The company has demonstrated many steps of its proprietary harvesting technologies on Earth. It worked with the Vermeer Corporation, an industrial equipment manufacturer in Iowa, to develop and test a full-scale prototype of an excavator. It can dig up 100 metric tons of dirt per hour and return it back to the surface in one swift motion.

Interlune successfully tested sorting out smaller particles in a mock lunar atmosphere on parabolic flight, which simulates the moon's lower gravitational pull. Lai said — in perhaps a first on Earth — they also separated Helium-3 from its Helium-4 mixture.

The firm is primarily focused on the harvesting technology once on the moon, leaving travel and transportation of its equipment there to other companies.

“It's really all about establishing a resilient supply chain for this critical material,” Meyerson said. He said retrieving Helium-3 for quantum-computing is the company's focus more imminently, but he could see other uses for the element in the long term, including fusion energy.

Other analysis has shown harvesting Helium-3 is challenging, if not unrealistic. A 2022 report from the U.S. Geological Survey classified the element as an “inferred unrecoverable resource” at the time, with the likelihood of such reserves present in 30 years as “unknown.” The retrieval may also require digging up a lot of lunar dirt to retrieve a few liters.

Interlune is aware of the challenges. Lai said the company is putting a multispectral camera on a commercial rover to the moon later this year, which will scout reserves of Helium-3.

Interlune has signed agreements to sell three liters of lunar Helium-3 to the U.S. Department of Energy Isotope Program, which distributes Helium-3 from the U.S. stockpile of tritium decays, in the first government purchase of a natural resource from space.  Commercial company Maybell Quantum has agreed to purchase thousands of liters of Helium-3.

Bluefors's more substantial commitment indicates a growing demand for the product in the near future, Lai said.

“Maybe three to five years, we are going to see data centers with quantum computers, and it will be on a lot larger scale than we are today,” said Waismaa, of Bluefors. “For that, we saw that we need to have several suppliers of Helium-3.”

What readers are saying

The comments express skepticism and concern about the potential mining of Helium-3 from the moon, with many questioning the legality and ethical implications of such endeavors. Several comments highlight the speculative nature of the venture, likening it to a Ponzi scheme or a... [Show more](#)

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